5. Find the domain of the function $f(x) = \sqrt{1+\sqrt{1-\sqrt{1-x^2}}}$

6. If $f(x) = x^2$, $g(x) = 3x$ and $h(x) = x-2$, Prove that $(f \circ g) \circ h = f \circ (g \circ h)$.

7. Let $A = \{1,2\}$ and $B = \{1,2,3,4\}$, $C = \{5,6\}$ and $D = \{5,6,7,8\}$. Verify whether A×C is a subset of B×D?

8. If $f(x) = \dfrac{x-1}{x+1}$, $x \neq -1$ show that $f(f(x)) = -\dfrac{1}{x}$, provided $x \neq 0$.

9. The functions f and g are defined by $f(x) = 6x + 8$; $g(x) = \dfrac{x-2}{3}$

   (i) Calculate the value of $gg\left(\dfrac{1}{2}\right)$

   (ii) Write an expression for $gf(x)$ in its simplest form.

10. Write the domain of the following real functions

    (i) $f(x) = \dfrac{2x+1}{x-9}$ (ii) $p(x) = \dfrac{-5}{4x^2+1}$ (iii) $g(x) = \sqrt{x-2}$ (iv) $h(x) = x+6$

Points to Remember

- The Cartesian Product of A with B is defined as $A \times B = \{(a,b) \mid \text{for all } a \in A, b \in B\}$
- A relation R from A to B is always a subset of $A \times B$. That is $R \subseteq A \times B$
- A relation R from X to Y is a function if for every $x \in X$ there exists only one $y \in Y$.
- A function can be represented by
  - (i) an arrow diagram
  - (ii) a tabular form
  - (iii) a set of ordered pairs
  - (iv) a graphical form
- Some types of functions
  - (i) One-one function
  - (ii) Onto function
  - (iii) Many-one function
  - (iv) Into function
- Identity function $f(x) = x$
- Reciprocal function $f(x) = \dfrac{1}{x}$

1 Relation and Functions.indd 34 15-12-2020 02:50:51 PM